Manisha Kaler

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Education

University of Massachusetts Jan 2026 - Dec 2027 (Expected)
Masters of Science in Computer Science 3.9/4.0
Coursework: Machine Learning, Performance Engineering, Data Science Fundamentals

Indian Institute of Technology Aug 2017 - May 2021
Bachelors of Technology in Computer Science and Engineering
Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Networks, Operating Systems, Computer Organization and Architecture, Algorithms Design and Analysis, Theory of Computation, Computer Graphics

Work Experience

Mercedes-Benz Research & Development, North America
Software Development Intern, MB Intelligence Context Team Aug 2026 – Present

- Built and deployed multi-region AWS infrastructure for **MB Canvas**, a customer-facing AI in-vehicle display personalization feature, across **AMAP**, **EMEA**, and China using **AWS CDK (Go)** and **GitLab CI/CD**.
- Built a cross-account **S3 asset-transfer pipeline** for AI-generated backgrounds, configuring S3 bucket policies, IAM permissions, and customer-managed **AWS KMS** key policies; designed and deployed **ComfyUI** workflows for image generation, resizing, visual-safety checks, DocumentDB storage, and delivery to regional accounts.

Arun Data Processing
Software Engineer 2 Feb 2023 - Dec 2025

- Built and shipped a **Go-based PDF generation microservice** for the Income Tax Department of India's production tax filing workflow, supporting **80M+ taxpayers**.
- Engineered **schema-driven rendering logic** that mapped nested JSON structures to reusable **Go templates** for tax forms, acknowledgements, and verification PDFs.
- Benchmarked third-party **HTML-to-PDF renderers** and selected a compatible solution to improve PDF generation reliability for legacy **RHEL Linux** infrastructure.
- Built a **schema-driven code generation pipeline** that generated Angular HTML, TypeScript, and service-layer code, reducing new workflow delivery time from **4 months to 3 weeks**.
- Developed **30+ Angular components** and shared modules for validation, localization, and **WCAG-aligned accessibility** across online and offline filing flows.
- Integrated **20+ client REST APIs** end-to-end and packaged the Angular utility into cross-platform desktop executables using **Wails**.

VISA
Software Engineer, Cloud PaaS Team Jul 2021 - Jan 2023

- Optimized ScalableRun, a **global script execution platform** managing **10,000+ servers**, improving platform reliability through **real-time validations** and **asynchronous processing**.
- Designed event-driven pub-sub systems using **message queues** to monitor **Ansible Tower** automation, surfacing execution metrics in **Grafana dashboards** to reduce **MTTR by 30%**.
- Implemented the frontend for **Ansible group management**, enabling users to organize nested server groups for bulk script execution and improving deployment speed by **20%**.
- Built an **IAM-based access validation layer** to enforce user permissions across server groups, ensuring secure script executions across global **Cloud PaaS** infrastructure.
- Built **unit and integration tests** for Spring Boot services using **JUnit**, **Mockito**, and Spring test utilities, achieving **95% code coverage**.

Software Engineer Intern May 2020 - Jul 2020

- Developed a full-stack web application using **Java** and **Visa Developer APIs** to provide **geo-targeted offer recommendations** for small merchants.
- Prototyped a functional **payroll API** solution in Java, digitizing merchant payment workflows for a global case challenge.

Project Experience

Lecture Notes AI Assistant ↗ | *LLM-based Study Tool* UMass Amherst

- Built an **LLM-powered** study assistant using **Streamlit** and **Gemini 2.5 Flash-Lite** to summarize transcripts, answer grounded questions, and generate study notes.
- Implemented a structured storage layer with **SQLAlchemy** and **SQLite** to persist transcripts and support retrieval grounded responses.

Cache-Optimized AVX Matrix Multiplication Kernel ↗ | *C++, AVX, OpenMP, Linux perf* UMass Amherst

- Optimized **C++ matrix multiplication** with loop reordering, cache blocking, **AVX vectorization**, register-aware kernels, and **OpenMP**; achieved **4.48x** single-thread and **2.44x** 8-thread speedup on large workloads.
- Used **Linux perf** and assembly inspection to analyze performance, reducing **L1 cache miss rate** from **72.47%** to **6.56%**, then to **1.25%** with a cache-aware tiled kernel.

Technical Skills

Languages: Go, Java, TypeScript, JavaScript, C/C++, SQL, HTML/CSS, Python
Frontend: Angular, React, Next.js, Bootstrap, Wails, Responsive Design
Backend & APIs: Spring Boot, Gin, RESTful APIs, Microservices
AI & LLM: Prompt Engineering, RAG Pipelines, Gemini API, Grounding
Cloud & DevOps: AWS (CDK, S3, IAM, KMS, DocumentDB, Route 53, VPC, EC2, RDS, ECS, API Gateway, Lambda), GitLab CI/CD, Docker, Jenkins, Ansible Tower, Linux, Git
Data & Tools: PostgreSQL, MySQL, MongoDB, Hazelcast, SQLite, SQLAlchemy, Grafana, perf
Testing & Process: Unit Testing, Integration Testing, Agile/Scrum